

- But holes labelled a, b, c, d and e are not connected to holes labelled f, g, h, i and j.
 - ♦ As can be seen in the image, there are 30 such groups on each side in the breadboard.
 - ♦ The groups of perforations beyond this group are connected in an exactly perpendicular fashion as shown by orange and yellow lines in the image.
- Any IC used in a circuit is placed in the middle part with either side of the legs of the IC on either side of the separating ravine as shown in the figure.

LED (Light Emitting Diode)

A ‘light emitting diode’ (LED) emits light when an electric voltage is applied across the ends of the LED.

When using an LED in a circuit, a resistor is connected along with it, to avoid damage to the LED due to high current.

Multimeter

A ‘multimeter’ is a device that measures multiple electrical parameters such as voltage, current and resistance. A digital multimeter (as shown in the image) displays the required value by selecting the required quantity using the dial below the display.



A digital multimeter

Continuity tester

After creating an electronic circuit on a breadboard or a printed circuit board (PCB), things may not work as expected. Though everything may seem fine, this happens if there are loose or faulty connections between two points. The continuity tester is used to check if there are any such problematic connections.

Please note that the continuity tester indicates whether a short circuit exists between any two points and isn't effective for checking whether there's resistance between two points. You can even make a continuity tester using an LED, a battery, a resistor and some wire to connect them. If you have access to a multimeter, there's an in-built continuity tester. It can be used by turning the knob to the symbol shown in the image. In the presence of a continuity, the device makes a buzzing sound.



Continuity tester symbol on a digital multimeter